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GENERAL PROJECT REVIEW CONSOLIDATED REPORT (HE)

COVER PAGE

PROJECT	
Project number:	101139292
Project acronym:	Opti-6G
Project name:	Optical 6G Cell Free Networks
Call:	HORIZON-JU-SNS-2023
Topic:	HORIZON-JU-SNS-2023-STREAM-B-01-03
Type of action:	HORIZON-JU-RIA
Service:	SNS
Project starting date:	1/1/2024
Project duration:	36 months

PROJECT REVIEW	
Period covered:	from 1/1/2024 to 30/6/2025
Reporting period number:	1st
Date of the latest version of DoA against which the assessment is performed:	19/9/2025
Date of meeting with consortium (if applicable):	1/10/2025 – 1/10/2025
Name of project officer:	Oana RADU
Name(s) of monitors:	- Laura COTTATELLUCCI - IULIUS VINCENTIU COJOCARU - GIANCARLO DE MARCHIS

1. OVERALL ASSESSMENT

1. Overall assessment

○ The project has fully achieved its objectives and milestones for the period.

◉ The project has achieved most of its objectives and milestones for the period, with relatively minor deviations.

○ The project has achieved some of its objectives and milestones. However, corrective action will be required.

○ The project has failed to achieve critical objectives and/or milestones and/or is severely delayed.

2. General comments (executive summary)

Opti-6G is a project focused on developing Optical 6G Cell Free Networks for improved indoor communications.
It utilizes a photonic near-infrared (IR) cell-free network, often involving Optical Wireless Communication (OWC) access points. The use of the optical unlicensed bands helps avoid interference and the need for frequency licenses. It aims to provide universal broadband coverage within buildings, achieve high data rates, and deliver high-precision localization (centimetre-level) for applications like Industry 4.0 and automated guided vehicles.
The OPTI-6G project has made substantial and technically sound progress during its first reporting period, successfully completing the foundational "Definition and Design Phase". The consortium has produced all key deliverables related to use cases, system architecture, and the initial designs for the communication and localisation systems. Dissemination and exploitation activities are a clear strength, with numerous scientific publications, two patent filings, and active engagement in key industry events. The project's objectives remain highly relevant, addressing ambitious topics like optical cell-free networks and high-precision localisation.
However, a consistent pattern of minor delays (from two weeks to nearly two months) was observed across almost all deliverables and the first milestone, indicating a potential issue in planning and timeline management. During the review the Project was asked if any of these delays are seen to have a negative impact on other activities and future plan, and they said that they don't expect any negative effects.
Overall, the project is on a solid technical trajectory, but sustained efforts are required to improve timeline adherence, KPI consistency, and resource management to ensure its continued success. Looking at the consortium's performance within this reporting period, the review panel believes that the project should successfully complete the actions listed in the GA, within the GA timeline.

3. Recommendations

The periodic report and deliverables D2.1, D2.2, and D3.1 were updated according to the following recommendations done during the review meeting:

1. Address KPI Misalignment: The consortium must urgently clarify the discrepancy between the target KPIs in WP2 and the achievable KPIs in WP3 and WP4. A revised plan should be presented to either enhance the system design to meet the targets or formally justify a revision of the initial KPIs.
2. Revise Key Deliverables: Revisions have been requested for D2.1, D2.2, and D3.1. The consortium is requested to address the points raised, specifically by providing a clearer justification for the target KPIs (D2.1) and a more detailed technical description and rationale for the chosen interface between the Optical Wireless Network (OWN) and the 5G Cell-Free Network (5G-CFN) (D2.2, D3.1).
3. Justify and Rebalance Resources: The consortium must provide a clear justification for the reported overspending and underspending by partners.
4. Strengthen Risk Management: The risk management process should be updated to be less optimistic and more proactive. It must explicitly include risks related to the KPI mismatch, hardware/software integration, and the reproducibility of experimental results.
5. Accelerate Standardisation Engagement: The project should move beyond monitoring and accelerate its engagement with standardisation bodies (e.g., IEEE, ITU, 3GPP) by preparing and submitting concrete technical contributions based on project results.

The following recommendations remain for the next period:
a. Provide a Recovery Plan: To address the consistent delays, the consortium should provide a recovery plan with revised internal deadlines for the second reporting period. This is crucial to prevent cascading delays in the more complex integration and validation phases.
b. A plan to rebalance partner effort should be implemented for the next period to be in line with the DoA.
c. Enhance Exploitation Plans: Building on the strong initial work, the exploitation plans should be enhanced with more specific, measurable business models, clear commitments for pilot projects, and quantified KPIs to translate technical work into tangible market impact

4. EIC follow up actions (for EIC actions)
Not applicable
5. Does the project meet the necessary conditions for receiving additional grants under the EIC? (for EIC actions)
○ Yes ○ No ◉ Not applicable
Not applicable

2. OBJECTIVES AND WORK PLAN

1. Is the progress reported in line with objectives and work plan as specified in the DoA? If there are significant deviations, please comment.

○ Yes
○ No
◉ Partially
○ Not applicable

Partially. The technical work is consistent with the DoA, and all required deliverables for the period have been submitted. However, progress is hampered by consistent delays in submissions and a resulting delay in achieving milestone MS1. A critical deviation is the emerging gap between the target KPIs defined in WP2 and the performance capabilities of the systems designed in WP3 and WP4.

- WP1 - Project Management: On track. Coordination has been effective, and management deliverables were submitted on time.
- WP2 - Use Cases, Requirements, and Architecture: Good progress, delivering technically strong documents defining use cases (D2.1) and architecture (D2.2). However, deliverables D2.2 and D2.3 were submitted with a one-month delay. Milestone MS1 was formally achieved, but late. A key concern raised during the review meeting was that the target KPIs in D2.1 lacked clear justification and appeared misaligned with what would be achievable later. The issue was addressed in the resubmitted deliverables
- WP3 - Near-IR Communication System Design: The deliverable D3.1 presents a detailed and ambitious system design. However, it was delivered with a significant delay of over one month. A more detailed description of the novel interface between the OWN and the 5G-CFN was needed, as the current design appeared potentially redundant or non-optimized. The Deliverable was resubmitted.
- WP4 - Localisation and Sensing System Design: The design in D4.1 is innovative, particularly its fusion of multiple localisation techniques. However, this was the most delayed deliverable (~1.5 months), putting pressure on subsequent integration tasks. Critically, the KPIs achievable with the proposed design do not satisfy the requirements defined in WP7.
- WP5 - Impact Creation: Progress is very strong. The project has an excellent record of dissemination, with numerous papers and active participation in major events. Exploitation plans are concrete and show strong potential. The primary area for improvement is to increase direct contributions to standardisation bodies.

2. Are the objectives of the project still scientifically, technologically and economically relevant?

◉ Yes
○ No
○ Partially
○ Not applicable

The project's objectives remain scientifically, technologically, and economically relevant, positioning OPTI-6G at the forefront of research on Optical Wireless Communication, cell-free networks, and high-precision localisation for Industry 4.0. These domains are of strategic importance for the development of 6G and carry substantial industrial and economic potential, as highlighted in deliverables D2.1 and D5.1. The integration of an Optical Wireless Network over a 5G Cell-Free Network represents a concrete and forward-looking step toward the long-term vision of a fully native optical cell-free system, combining the precision of optical channels with the flexibility of 5G architectures. The objectives and methodology outlined in the Description of Action remain achievable within the available resources and timeframe, provided that critical risks continue to be managed effectively. Minor adjustments and clarifications to objectives and TRLs, as reflected in the periodic report, are acceptable insofar as they do not alter the project's scope, although any rewording of technical objectives should be validated in review discussions to ensure alignment with the agreed DoA.

3. Critical risks

○ Yes
○ No
◉ Partially
○ Not applicable

The consortium maintains a risk register and proactively identified an unforeseen geopolitical risk. However, the

overall risk assessment is viewed as overly optimistic. Risks related to hardware/software integration and experimental reproducibility also require more active tracking.

4. Does the project respect the 'do no significant harm' principle?

- ◉ Yes
- ○ No
- ○ Partially
- ○ Not applicable

The project's activities are centred on communication technologies and do not involve activities that would violate this principle. The ethical considerations in D1.1 confirm this alignment.

5. Is the gender dimension appropriately taken into account?

- ○ Yes
- ○ No
- ○ Partially
- ◉ Not applicable

The project addresses gender matters in D1.1 by committing to monitoring gender balance in staffing. Several women are listed as authors and contributors across the deliverables, including in leadership roles for dissemination activities. But, actually, the gender dimension deals with the objective of the research and not with gender balance and the issue of gender dimension does not appear relevant for this particular project.

6. Does the project respect the commitments concerning open science as described in the DoA? Is it undertaking additional open science practices?

- ◉ Yes
- ○ No
- ○ Partially
- ○ Not applicable

The project demonstrates a consistent and proactive commitment to open science. The consortium adheres to the principles and practices defined in the Data Management Plan (D1.1) and in the Description of Action, ensuring that research outputs are made as open as possible and as closed as necessary. Trusted repositories such as Zenodo are used for data storage and dissemination, with DOIs assigned to enhance findability. A significant number of deliverables and publications are already publicly accessible, and clear policies have been established to guarantee that these outputs remain available beyond the project's lifetime. Furthermore, the consortium has made available the datasets used for training deep neural networks, reinforcing transparency and the long-term usability of project results.

7. Is the project adequately integrating social sciences or/and humanities? (for SSH topics)

- ○ Yes
- ○ No
- ○ Partially
- ◉ Not applicable

Non applicable

8. Have the ethics/security deliverables due for the current period been adequately addressed and approved?

- ◉ Yes
- ○ No
- ○ Partially
- ○ Not applicable

Deliverable D1.1, "Ethics and Data Management Action Plan Report," was submitted and adequately addresses all relevant ethics and data management aspects, including GDPR compliance and handling of research data.

9. Did the fellows/staff members demonstrate sufficient knowledge of the research project? (for Marie Skłodowska Curie actions)

○ Yes ○ No ○ Partially ◉ Not applicable
Not applicable
10. Were the fellows aware of their rights and obligations as a Marie Skłodowska Curie fellow? (for Marie Skłodowska Curie actions)
○ Yes ○ No ○ Partially ◉ Not applicable
Not applicable
11. Did any issues requiring REA follow-up arise during the meeting? (for Marie Skłodowska Curie actions)
○ Yes ○ No ○ Partially ◉ Not applicable
Not applicable
12. Has the project effectively addressed the relevant standardisation aspects in R&I activities?
○ Yes ○ No ○ Partially ◉ Not applicable
Not applicable
13. Did the EIC beneficiaries prioritise IP protection over dissemination when applicable, particularly with regards to results with market potential?
○ Yes ○ No ○ Partially ◉ Not applicable
Not applicable
14. Have the comments and recommendations from previous project reviews been taken into account?
○ Yes ○ No ○ Partially ◉ Not applicable
This is the first review

3. IMPACT

1. Is the proposed pathway to achieve the expected outcomes and impacts still credible?

- ◉ Yes
- ○ No
- ○ Partially
- ○ Not applicable

The project's pathway to impact is overall credible and well substantiated. The consortium has established a proactive dissemination and engagement strategy, supported by the activities described in D5.2 and by a strong presence in major industrial and scientific events such as MWC and EuCNC. The techno-economic analysis presented in D5.1 confirms the project's potential to contribute to Industry 4.0 markets, while continued participation in SNS JU working groups reinforces its visibility and relevance. The technical objective of implementing an OWN over a 5G-CFN remains fully credible, even if some KPIs may not yet be completely achieved, as these are considered necessary preliminary steps. However, exploitation plans would benefit from further detail and from more tangible industrial commitments to ensure that the expected impacts are effectively realised.

2. How will the project have an impact on policy making (if any)?

- ○ Yes
- ○ No
- ◉ Partially
- ○ Not applicable

The OPTI-6G project, while primarily a technology-driven research and innovation initiative focused on optical cell-free networks and localisation technologies for 6G, may indirectly contribute to policy-making through its alignment with European strategic priorities. Its work on optical wireless communications, localisation, and integration with cell-free RANs supports the objectives of the EU's Smart Networks and Services Joint Undertaking and the broader European 6G strategy. The project's dissemination and liaison activities (D5.2) already demonstrate coordination with major standardisation bodies such as IEEE, ITU, 3GPP, and LCA, thereby contributing to future regulatory frameworks on spectrum use, safety, and interoperability. In addition, the techno-economic analysis (D5.1) connects the project's results with EU industrial and digital-transformation policies, particularly regarding Industry 4.0 and energy efficiency—both relevant to the Green Deal and the digitalisation agenda. While direct influence on policy development or implementation may remain limited at this stage, the project provides valuable technical evidence that could inform future EU and international regulatory and standardisation processes related to OWC technologies and 6G integration.

3. Please indicate the EU headline priorities to which the project has contributed or will contribute to.

- ☐ European Green Deal
- ☐ Economy that works for people
- ☑ Europe for the digital age
- ☐ Promoting the European way of life
- ☐ A stronger Europe in the world
- ☐ A new push for democracy

The project directly contributes to this priority by developing advanced 6G networking technologies foundational for Europe's digital sovereignty. There is also potential relevance to the "European Green Deal," but this needs to be substantiated with evidence of, for example, improved energy efficiency.

4. Are the measures to maximise impact still suitable? (n/a for EIC Pathfinder)

- ◉ Yes
- ○ No
- ○ Partially
- ○ Not applicable

The measures to maximise impact are appropriate, well structured, and actively implemented. The consortium has

developed a comprehensive dissemination, communication, and exploitation strategy, supported by an effective online presence and strong participation in international conferences and events. Engagement with the SNS JU Architecture and Test/Measurement working groups ensures alignment with the broader European 6G vision and strengthens the project's contribution to the European research and innovation landscape. A solid publication record and patent-oriented exploitation plans further demonstrate commitment to visibility and knowledge transfer. Nonetheless, opportunities for improvement remain in strengthening direct participation in standardisation activities, as involvement so far has been mostly limited to monitoring rather than active contribution. Overall, the consortium's approach effectively supports the translation of project results into tangible scientific, industrial, and strategic impact.

5. Are the measures for public/stakeholder engagement properly implemented? (for EIC Pathfinder)

○ Yes
○ No
○ Partially
◉ Not applicable

Not applicable

6. Translation into innovation (for EIC Pathfinder)

○ Yes
○ No
○ Partially
◉ Not applicable

Not applicable

7. Empowering key actors (for EIC Pathfinder)

○ Yes
○ No
○ Partially
◉ Not applicable

Not applicable

4. IMPLEMENTATION

1. Has the project been efficiently and effectively managed (including risk management)?

○ Yes
○ No
◉ Partially
○ Not applicable

Project management is overall adequate and effective, with EURESCOM providing structured coordination, regular consortium meetings, and appropriate quality assurance mechanisms. The consortium has demonstrated strong collaboration and timely communication, ensuring alignment with the Description of Action and maintaining project coherence. Most deliverables—eight in total for the first reporting period—were submitted on time and of good quality, including the well-prepared Ethics and Data Management Action Plan (D1.1). However, recurring delays across several deliverables point to potential weaknesses in timeline management or the original planning assumptions that should be addressed moving forward. The progress report was slightly delayed, and some rewording of technical objectives within it should be formally discussed at the review to maintain full consistency with the DoA. Apart from some budget-related uncertainties, there is no indication that the project has not been managed efficiently or effectively. Strengthening schedule control and ensuring timely justification of minor deviations would further enhance the robustness of project coordination.

2. Have all the obligations described in the grant agreement (contract) been respected by the participants (including ethics and security requirements, if applicable)?

◉ Yes
○ No
○ Partially
○ Not applicable

The consortium has complied with the obligations established in the Grant Agreement, fulfilling all contractual deliverables for the first reporting period, including the Ethics and Data Management Action Plan (D1.1), which satisfactorily addresses ethics, data protection, and security aspects. EU visibility requirements have been properly implemented across deliverables and dissemination materials. While a few deliverables—such as D3.1 and D4.1—were submitted later than planned, these delays did not have a significant impact on overall project progress. The project's management and reporting practices are consistent with the beneficiaries' responsibilities under the Grant Agreement. No ethical or security non-compliance has been identified, though future risk reporting would benefit from a more realistic assessment of integration challenges.

3. Have all participants contributed to the project according to the work-plan described in the DoA?

○ Yes
○ No
◉ Partially
○ Not applicable

All partners are contributing actively to the project's deliverables. However, resource usage has been uneven. According to the provided financial models for the use of the resources, there is a strong over-contribution from REL, justified as strategic front-loading, and a significant under-contribution from UVSQ and Oledcomm. A rebalancing of effort is required for the next period to ensure all tasks are adequately resourced according to the plan.

4. Have the EIC beneficiaries protected the IPR of the results with market potential (foreground) as planned in the DoA (including patents filing or any other formal IPR protection)? How do the beneficiaries plan to get return on investment over the generated IP? (for EIC Pathfinder)

○ Yes
○ No
○ Partially
◉ Not applicable

Not applicable

5. Have the participants disseminated project results (foreground) and have they communicated project activities and results as planned in the DoA (e.g. through publications, a page for the project on social media, press-releases, a website, video/film, etc.) and have they included a reference to EU funding?

- ◉ Yes
- ○ No
- ○ Partially
- ○ Not applicable

Dissemination and communication activities are excellent and a major strength of the project. The consortium has established a professional online presence, published an impressive number of scientific papers (7 conference, 6 journal), filed two patents, organized workshops, and actively participated in key industry events, always acknowledging EU funding.

6. If the plan for exploitation and dissemination provides for exploitation primarily in non-associated third countries, have the participants explained how that exploitation is still in the EU interest? Is it acceptable?

- ○ Yes
- ○ No
- ○ Partially
- ◉ Not applicable

Not applicable

7. Are the critical implementation risks and mitigation actions described in the DoA still relevant?

- ◉ Yes
- ○ No
- ○ Partially
- ○ Not applicable

The project's risk management framework remains relevant and generally effective, with the consortium demonstrating a proactive approach in identifying and mitigating both existing and emerging risks. The risks defined in the Description of Action continue to apply, and a new geopolitical risk has been recognised and addressed with appropriate mitigation actions. Nonetheless, risk reporting tends to be overly optimistic and would benefit from more critical and quantitative assessments to better reflect potential impacts and residual exposure. In addition, while the risk of not fully achieving certain target KPIs persists, continued technical progress and iterative refinement of the system are expected to bring the project closer to meeting the performance levels required for its key use cases. Strengthening the rigour and transparency of risk tracking in the next reporting period would further enhance project resilience and credibility.

5. RESOURCES (N/A FOR LUMP SUM AND UNIT GRANTS)

1. Were the resources used as described in the DoA and were they necessary to achieve the project objectives?

- ○ Yes
- ○ No
- ◉ Partially
- ○ Not applicable

The use of resources is generally consistent with the project's objectives and the principles of sound financial management, though some imbalances are evident. Financial execution currently stands at around 66% of the total grant after approximately 42% of the project's duration, suggesting a faster-than-expected expenditure rate that could lead to compression of activities and resource constraints in the second reporting period. While overall spending aligns with the Description of Action and the consortium has managed funds prudently, person-month utilisation is lower than planned, and deviations across partners highlight the need for closer monitoring and corrective planning. A comprehensive review of budget execution and workload distribution is therefore recommended to maintain coherence between technical progress and financial performance.

2. If there are significant deviations from planned budget, have they been satisfactorily justified?

- ○ Yes
- ○ No
- ◉ Partially
- ○ Not applicable

The consortium has provided explanations for the identified deviations in resource utilisation, which appear generally plausible but indicate the need for improved financial planning. REL's substantial overspend on person-months is justified as a strategic front-loading of work to deliver key components early and mitigate project risks—a proactive approach that supports timely progress. Conversely, UVSQ's significant underspend, attributed to coverage of an expert's effort from alternative sources, raises some concerns regarding the accuracy of the initial budget allocation and the consistency of resource availability for future periods. While only a few deviations have been reported, and these have been partially justified, a more rigorous review and rebalancing of resource planning will be necessary in the next reporting period to ensure alignment between work performed, financial execution, and project objectives.

3. If unforeseen subcontracting costs are declared, do you agree with them?

- ○ Yes
- ○ No
- ○ Partially
- ◉ Not applicable

Not applicable

4. If unforeseen in-kind contributions costs are declared, do you agree with them?

- ○ Yes
- ○ No
- ○ Partially
- ◉ Not applicable

Not applicable

Annex 1

Expert opinion on deliverables

Deliverable number	Deliverable name	Status	Comments
D1.1	Ethics and Data Management Action Plan Report	Accepted	A complete and adequate plan covering all required aspects. Submitted with a minor delay.
D2.1	OWC Cell Free Network Use Cases and User, functional and Technical Requirements	Accepted	The document is comprehensive and defines a relevant scope based on Industry 4.0 use cases. Revision was requested to provide a clearer technical justification for the absolute values of the target KPIs, as their origin was unclear and they appeared misaligned with the achievable performance of the proposed system.
D2.2	OWC Cell Free Network Architecture	Accepted	A detailed technical architecture that serves as a central design document. Revision was requested to provide a more detailed explanation and justification for the novel interface between the 5G radio network and the OWC network. The current description was unclear and suggested the design might be redundant or not fully optimized. The initial version was submitted with a ~1-month delay.
D2.3	OWC Cell Free Network Testing and KPI Verification Methodology	Accepted	A methodologically solid plan for testing and verification. It was noted that the defined KPIs seem less ambitious than the state-of-the-art, likely due to reliance on existing components. Submitted with a ~1-month delay.
D3.1	Near IR Cell Free Communication System Design	Accepted	An ambitious and technically detailed design for the core communication system. Revision was requested for the same reason as D2.2: to clarify the OWC/5G-CFN interface design, which appeared redundant. Minor editorial issues also needed correction. This deliverable had a notable delay of over 1 month.
D4.1	OWC Localisation Sensing System Design	Accepted	An innovative and technically strong design, particularly the fusion of multiple localisation techniques. However, this was the most delayed deliverable of the period (~1.5 months), putting pressure on future tasks. It was noted that the target KPIs do not seem to improve with the more complex fusion methods.
D5.1	Technoeconomic Analysis and Exploitation Plans	Accepted	A good report with valuable market insight and a solid SWOT analysis. The exploitation plans are understandably preliminary but provide a concrete outline from each partner. Some confusion was noted between the titles and content of D5.1 and D5.2 that should be clarified.
D5.2	Report on dissemination and liaisons with regulatory, industry associations and standardization bodies activities	Accepted	An excellent and comprehensive report showcasing a very strong dissemination and communication record, with numerous publications and active event participation.

Annex 2

Expert opinion on milestones

Milestone number	Milestone name	Achieved	Comments
MS1	Completion of Definition and Design Phase	Yes	The milestone has been formally achieved, as all associated deliverables (D2.1, D2.2, D3.1, D4.1, and D5.1) have been submitted. However, it is critical to note that the milestone was met with a delay. The due date was 31/12/2024, but the milestone's actual completion date is recorded as 11/02/2025. This delay is a direct consequence of the delays observed across all the deliverables listed as its means of verification. While the technical content of the design phase is comprehensive, the delay highlights the timeline management issues noted in the overall assessment.